Unit I - Laser and Optical Fiber

Syllabus

Interaction of radiation with matter, Population Inversion and Optical resonance cavity, three and four level laser, Ruby laser, He-Ne laser, Properties and engineering applications of LASER

Optical fibers: Propagation by total internal reflection, structure and classification (based on material, refractive index and number of modes), Modes of propagation in fiber, Acceptance angle, Numerical aperture, Attenuation and dispersion. Applications of Optical fiber

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Laser

Introduction

Laser is the greatest technical invention of 20th century. Laser is an acronym of **"Light Amplification by Stimulated Emission of Radiation".** Laser has lots of application in our everyday life. It is used in Bar Code reading, Laser printing, Optical Communication, Eye Surgery, Dental Drilling, for cutting welding and drilling in industry, for cutting and polishing in diamond industry etc.

(Characteristics/Properties of Laser Light)

- Laser is largely different from ordinary light. Ordinary Light spreads all over the space but laser light does not spreads at all, it is **highly directional**.
- Ordinary light is made up of large number of wavelengths but Laser light is a single wavelength i.e. it is a **monochromatic** source of light.
- In Ordinary light all the wavelengths are out of phase but in laser light all the wavelengths are in phase, that is laser light is highly **coherent.**
- And also ordinary light is of very low intensity but laser light has **high intensity**.

Difference between Ordinary light and Laser light

Ordinary light	Laser light
1) Ordinary Light spreads all over the space hence it is non directional	1) laser light does not spreads at all, it is **highly directional**
2) Ordinary light is made up of large number of wavelengths hence it is of white colour	2) Laser light is a single wavelength i.e. it is a **monochromatic** source of light. It can be of red, green or pink colour
3) In Ordinary light all the wavelengths are out of phase	3) In laser light all the wavelengths are in phase, that is laser light is highly **coherent.**
4) Ordinary light is of very low intensity	4) laser light has **high intensity**

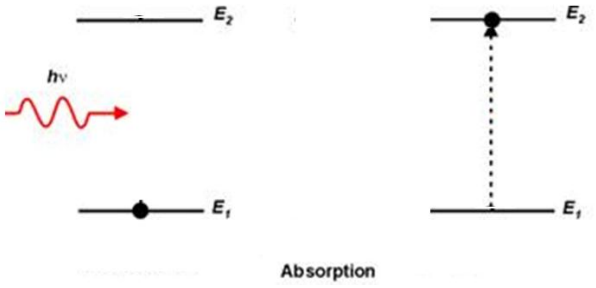
Interaction of light with matter and the three quantum process

An atom can move from one energy state to another energy state either by absorbing energy or by releasing the absorbed energy, equal to the energy difference between two energy states. It is termed as quantum jump or transition. The energy is absorbed or released in the form of photons.

The radiations of frequency ν incident on a medium, may be viewed as a stream of photons and each photon carrying $E = h\nu$. Then the interaction of radiations with atom leads to the following three distinct process in the medium.

Absorption

When a radiation of energy hν is incident on matter, atom absorbs energy and gets excited. The atom present in the lower energy state E_1 jumps to the excited state E_2 by absorbing a photon of energy hν, where $h\nu = E_2 - E_1$. This transition is known as absorption.



It may be represented as $A + h\nu \longrightarrow A^*$

Where A is an atom in lower energy state and A* is the atom in excited state

The number of absorption transition occurring in the material at any instant will be proportional to

(i) number of atoms N_1 in the lower energy state E_1
(ii) and the number of photons in the incident beam i.e., photon density $\rho(\nu)$

Thus the number of atom N_{ab} excited during the time Δt is therefore given by

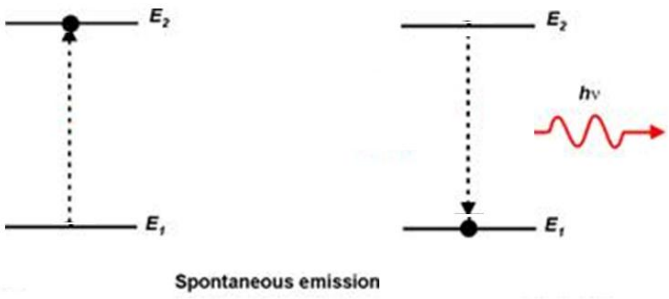
$$N_{ab} = B_{12} N_1 \rho(\nu) \Delta t$$

Where B_{12} is the probability of an absorption transition, it is also known as the Einstein coefficient for the induced absorption.

Emission

Spontaneous Emission

When energy radiations are incident on matter atom absorbs energy and gets excited to its higher energy state. The atom cannot remain in the excited state for longer period of time. It will come back to its ground state in a very short interval of time nearly 10^{-8} s, this is called as life time of an excited atom. So after completing its lifetime the atom returns back to its ground state on its own without any external impetus by releasing one photon of energy hν, where $h\nu = E_2 - E_1$. This is called as spontaneous emission.



It may be represented as $A^* \longrightarrow A + h\nu$

Where A is an atom in lower energy state and A* is the atom in excited state

The number of spontaneous transition occurring in the material at any instant will be proportional to

(i) number of atoms N_2 in the higher energy state E_2

Thus the number of spontaneous transition N_{sp} excited during the time Δt is therefore given by

$$N_{sp} = A_{21} N_2 \Delta t$$

Where A_{21} is the probability of a spontaneous transition, it is also known as the Einstein coefficient for the spontaneous transition.

Stimulated Emission

When energy radiations are incident on matter atom absorbs energy and gets excited to its higher energy state. The atom in the excited state, before it comes back to the ground state, is triggered with one more photon of energy $h\nu = E_2 - E_1$and made to jump to the ground state by emitting two photons which are identical in all respect with each other and are emitted in same direction and are highly coherent. This phenomenon of forced emission of photons is called as stimulated emission.

Stimulated emission

It may be represented as $A^* + h\nu \longrightarrow A + 2h\nu$

Where A is an atom in lower energy state and A* is the atom in excited state

The number of stimulated transition occurring in the material at any instant will be proportional to

(i) number of atoms N_2 in the higher energy state E_2
(ii) and the number of photons in the incident beam i.e., photon density $\rho(\nu)$

Thus the number of stimulated transition N_{st} occuring during the time Δt is therefore given by

$$N_{st} = B_{21} N_2 \rho(\nu) \Delta t$$

Where B_{21} is the probability of a stimulated transition is a constant of proportionality.

The emitted photons are identical in all respect with the incident photon. They have same frequency and are in phase with the incident photon. Both the photons travel in the same direction.

Difference between Spontaneous emission and stimulated emission

Spontaneous Emission	Stimulated Emission
1) Spontaneous emission is a random process	1) Stimulated emission is not a random process
2) This process cannot be controlled	2) This process is a controlled process
3) Photons are emitted uniformly in all directions and hence the light emitted is non directional	3)The photons are emitted in the same direction as that of the stimulating photon and hence the light produced is highly directional
4) Photons of slightly different frequencies are generated and hence the light is polychromatic	4) The photons emitted are of same frequencies and hence the light produced is monochromatic
5) Photons emitted do not have any correlation in their phases, therefore the light produced by this process is incoherent	5) The photons emitted by this process are all in phase and therefore, light is coherent

6) In this process multiplication of photon does not takes place, hence there is no amplification of light	6) In this process multiplication of photon takes place, and hence light amplification occurs
7) The light produced is of low intensity. Total intensity is given by $I_T = N\,I$ Where N is number of atoms and I is the intensity of each photon	7) The light produced is of high intensity. Total intensity is given by $I_T = N^2\,I$

Condition for light amplification

1) In order to enhance the number of stimulated transitions the radiation density ρ(ν) is to be made larger.
2) To increase the probability of the stimulated emissions, the life time of the atoms in the excited state should be larger. This is called as metastable state.
3) The stimulated emission will be larger than the absorption only when $N_2 > N_1$. This is called as population inversion and the light will be amplified. But if $N_1 > N_2$ the medium will absorb the light instead of amplifying it.

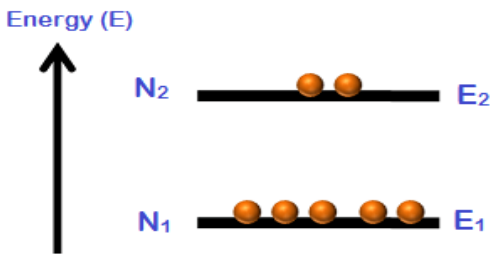
Population Inversion

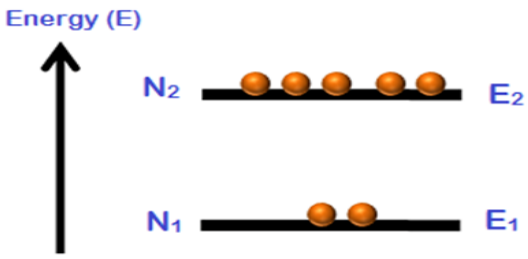
The number of atoms occupying an energy state is called as population of that state. In a state of thermal equilibrium there are more atoms in the lower energy level than in the upper energy level and the population ratio is governed by the equation

$$\frac{N_2}{N_1} = e^{-\left[\frac{\Delta E}{kT}\right]}$$

Where $\Delta E = E_2 - E_1$, and k is the Boltzmann constant

In order to achieve stimulated emission exclusively, there must be more atoms in the upper energy level than in the lower energy level. To make the number of atoms in the excited state more than the number of atom in lower energy state is called as population inversion





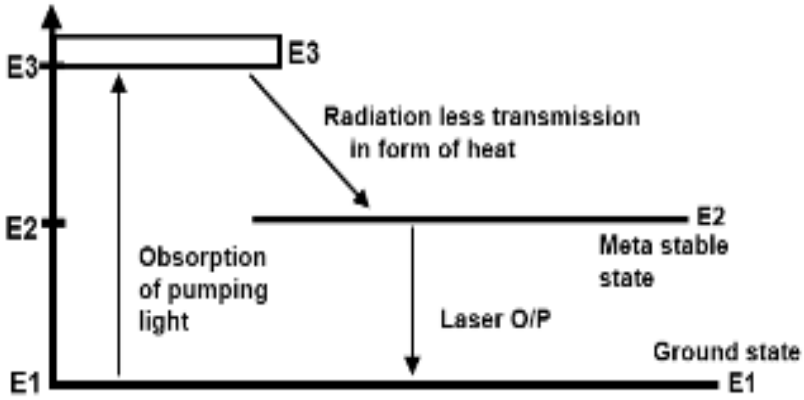
Metastable State

In order to achieve the condition of population inversion, the excited atoms are required to remain in the excited state for longer period of time so that large number of atoms gets accumulated. If the atom in the excited state remains in the excited state for longer period of time which is of the order of 10^{-6} to 10^{-3} s then it is said to be in its metastable state.

Pumping

In order to achieve the condition of population inversion the atoms are continuously raised from their lower energy level to higher energy level. Pumping is the process of supplying energy to raise the atoms in their lower energy level to the higher energy level. A lot of pumping methods are available like: **Optical pumping**: - by exposing the material to light radiations. **Electrical pumping**: - by producing electrical discharge which causes the ionization of the medium. **Chemical pumping:** - by chemical reaction

Three Level Pumping Scheme



- A typical three-level pumping scheme is shown in Fig. The state E_1 is the ground state, E_3 is the pump state and E_2 is upper lasing level, which is a metastable state.
- When the medium is exposed to pump frequency radiation, a large number of atoms are excited to pumping level E_3. But E_3 is highly unstable state so the atoms rapidly undergo downward transition to level E_2 through a non radiative transition.
- E_2 is the metastable state and hence there will be large accumulation of atoms at this level.
- When more than half of the ground state atoms accumulate E_2, the population inversion condition is achieved between the two states E_1 and E_2.
- Now a spontaneously emitted photon can trigger the stimulated emission of atom at E_2 to E_1 and a laser beam is obtained.
- In three level pumping scheme, the laser transition occurs between E_2 level and the ground state. Therefore, population inversion is achieved only when more than half of the ground state atoms are pumped to the upper state. Thus the scheme requires very high pump power.
- The three level pumping scheme produces light only in pulse mode

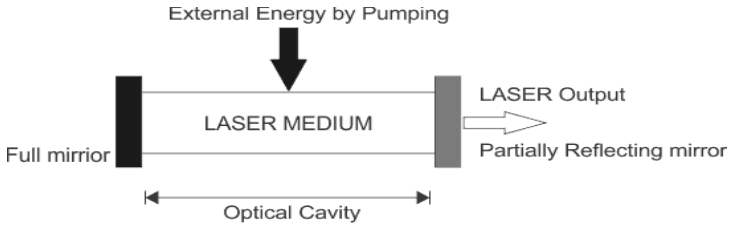
Four Level Pumping Scheme

- The typical four level pumping scheme is shown in Fig. The state E_1 is the ground state, E_4 the pumping level, E_3 the metastable upper lasing level and E_2 the lower lasing level. E_2, E_3 and E_4 are the excited states.
- When the medium is exposed to pump frequency radiation, a large number of atoms are excited to pumping level E_4. But E_4 is highly unstable state so the atoms rapidly undergo downward transition to level E_3 through a non-radiation transition.
- The level E_3 is the metastable state, the downward transition of atoms from E_3 to E_2 cannot takes place and hence population at E_3 grows rapidly.
- As a result, the level E_2 is virtually empty. Therefore, population inversion is attained between the states E_3 and E_2.

- Now a spontaneously emitted photon can trigger the stimulated emission of atom at E3 to E2 and a laser beam is obtained.
- From state E2 the atoms undergo a non-radiation transition to the ground state E1 and are again available for excitation making it possible for light to be continuously emitted.
- The lower laser transition level in this scheme is nearly vacant. Therefore, less pump power is sufficient to achieve population inversion.

Components of Laser

The components of laser are (i) Active Medium, (ii) a pumping agent (iii) and an optical resonator

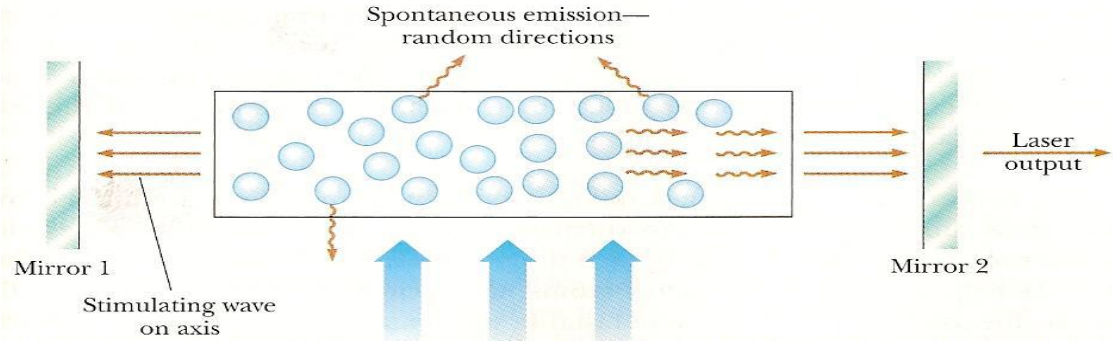


(i) **Active Medium**: The active medium is the material in which the laser action takes place. The active medium hosts the active centers. Active centers are the atoms which causes laser action.

(ii) **The Pump**: In order to achieve the condition of population inversion the atoms are continuously raised from their lower energy level to higher energy level. Pumping is the process of supplying energy to raise the atoms in their lower energy level to the higher energy level.

(iii) **Optical Resonator:** A pair of plane parallel reflecting mirrors, enclosing laser medium in between them is known as an optical resonant cavity. One of these mirrors is partially reflecting and the other is made fully reflecting.

Lasing action (working of optical resonant cavity)



Amplification

- A majority of photons traveling along the axis cause stimulated emission and are reflected back on reaching the end mirror.
- They travel towards the opposite mirror and on their way stimulate more and more atoms and build up the photon strength.
- The photons that strike the opposite mirror are reflected once more into the medium.The photons travel once more through the medium generating more photons and more amplification.
- The photons are then reflected again at the mirror and travel through the medium. As the photons are reflected back and forth between the mirrors, stimulated emission sharply increases and the amplification of light takes place.
- The mirror thus provides *positive feedback* of light into the medium so that stimulated emission acts are sustained and the medium operates as an oscillator.

Oscillations

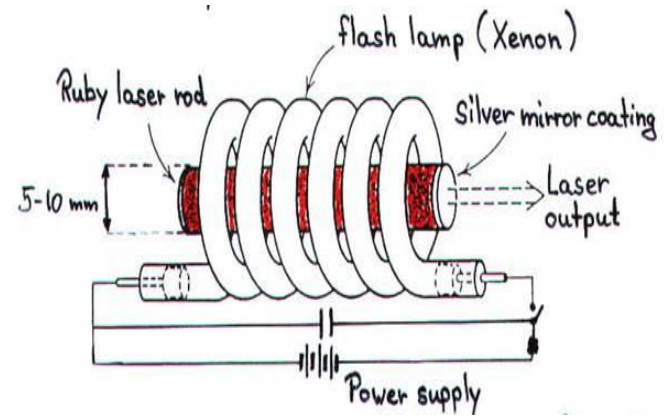
- At each reflection at the front-end mirror, light is partially transmitted through it.

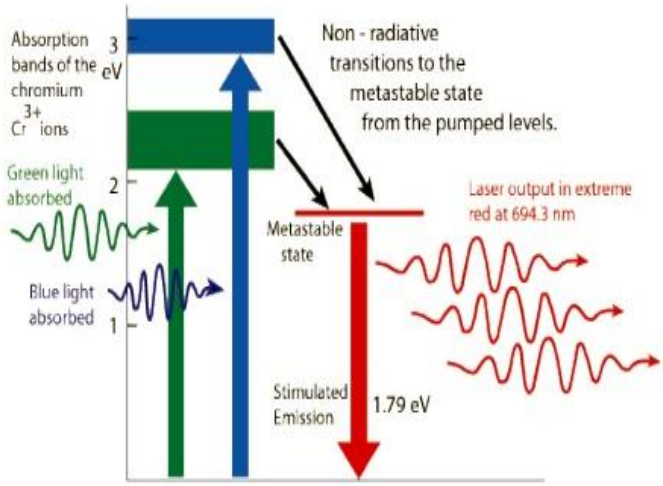
- The transmitted component constitutes a loss of energy from the resonator.
- When the losses at the mirror and within the medium balance the gain, a steady and strong laser beam will emerge from the front-end mirror.

Ruby Laser

Construction

- Ruby material is taken in the form of a cylindrical rod of about 4 cm in length and 1 cm in diameter. Its ends are grounded and polished such that the end faces are exactly parallel and are also perpendicular to the axis of the rod. T
- The end faces of the ruby rod are silvered so that they form the optical resonator. The rear face is made totally reflecting while the front face is made partially reflecting.
- The laser rod is surrounded by a helical photographic flash lamp filled with Xenon. Whenever activated by the power supply the lamp produces flashes of white light.





Working

- Ruby laser uses a three-level pumping scheme.
- The energy levels of Cr^{3+} ions in the crystal lattice as shown in Fig.
- The xenon discharge generates an intense burst of white light lasting for a few milliseconds. The Cr^{3+} ions are excited to the energy bands E_3 and E_3' by the green and blue components of white light.
- From there Cr^{3+} ions undergo non-radiative transitions and quickly drop to the metastable level E_2.
- The metastable state has a lifetime of approximately 1000 times more than the lifetime of E_3 level. Therefore, Cr^{3+} ions accumulate at E_2 level.
- When more than half of the Cr^{3+} ion population accumulates at E_2 level, the state of population inversion is established between E_2 and E_1 levels.
- A chance photon emitted spontaneously by a Cr^{3+} ion initiates a chain of stimulated emissions by other Cr^{3+} ions in the metastable state.
- Red photons of wavelength 6943Å travelling along the axis of the ruby rod are repeatedly reflected at the end mirrors and light amplification takes place. A strong intense beam of red light (λ = 6943Å) emerges out of the front-end mirror.
- The xenon flash lasts for a few milliseconds. However the laser does not operate throughout this period. Once stimulated transitions commence, the metastable state E_2 gets populated very rapidly and lasing ceases. The laser becomes inactive till the population inversion is once again established. Therefore, the output of the laser is not continuous but occurs in the form of pulse of microsecond duration.

Limitations

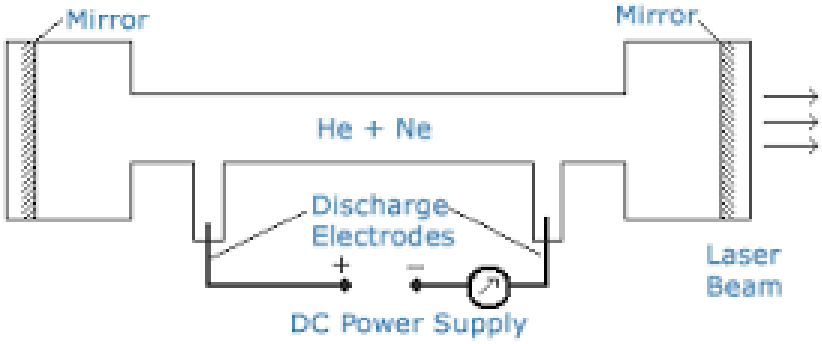
- The efficiency of ruby laser is very less as only the green component of the pumping light is utilized

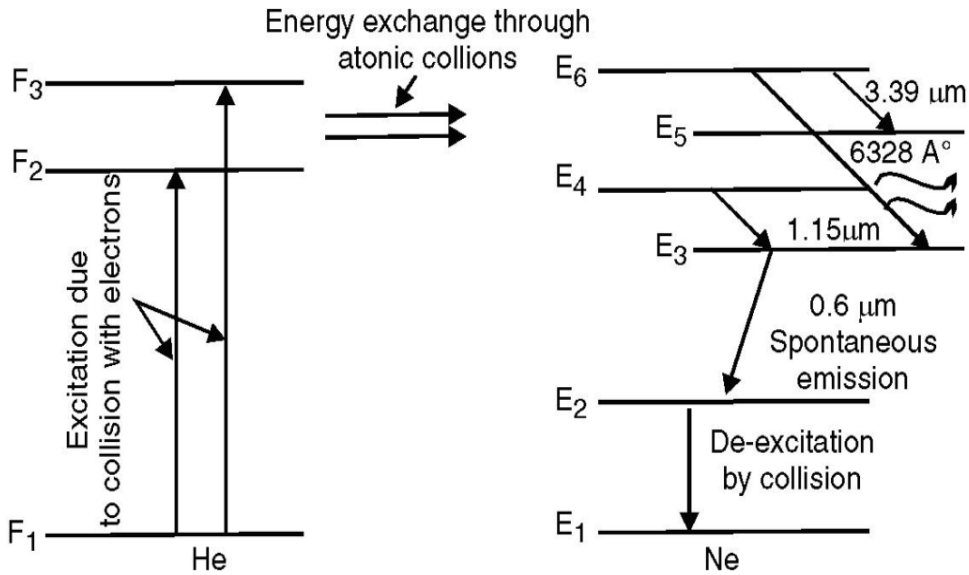
while the rest of the components of incident light are left unused.

- Secondly, it requires greater excitation energy as more than half of the active centers are to be lifted to the excited state from the ground state, in order to achieve population inversion between the two states.
- The output of the laser is not continuous but occurs in the form of pulse of microsecond duration.

He-Ne Laser

- The schematic of a He-Ne laser is shown in Fig. Helium-Neon laser consists of a long discharge tube filled with a mixture of helium and neon gases in the ratio 10:1. Flat glass quartz plates which function as a window are sealed to the tube at both ends. Two optically plane glass mirrors are fixed on either side of the tube normal to its axis which forms the optical resonator cavity.





Working

- Helium-Neon laser employs a four-level pumping scheme. The energy levels of helium and neon are shown in Fig.
- When power is switched on, a high voltage of about 10kV is applied across the gas. It is sufficient to ionize the gas.
- The electrons and ions produced in the process of discharge are accelerated towards the anode and cathode respectively.
- The energetic electrons excite helium atoms being lighter, through collisions to the excited levels F_2 and F_3 which lie at 19.81eV and 20.16eV above the ground state respectively.
- These two states are metastable states and the excited helium atoms cannot return to the ground state by spontaneously emitting photons of energy.
- However, it can return to the ground level by transferring its excess energy to a neon atom through collision. Such an energy transfer can take place when the two colliding atoms have identical energy

levels.

- The E_6 level which is at 20.66 eV and E_4 level which at 18.7 eV of neon atom nearly coincide in energy with the F_3 and F_2 levels of helium atom.
- Therefore, neon atoms acquire energy and go to the excited state on collision with helium atoms, which return to the ground state after transferring the energy.
- The kinetic energy of helium atoms provides the additional 0.05eV for exciting the neon atom to E_6 and E_4 level. This is the main pumping mechanism in He-Ne system.
- Neon atoms are the active centers and the role of helium is to excite neon atoms and cause population inversion.
- The probability of energy transfer from helium atoms to neon atom is more as there are 10 helium atoms to 1 neon atom in the mixture. For this reason, the probability of reverse transfer of energy, i.e., from neon to helium atom is extremely small.
- The E_6 and E_4 levels of neon atoms are also metastable states. Therefore population of Neon atoms in E_6 and E_4 increases due to successive collisions.
- At ordinary temperatures, the E_5 and E_3 levels of neon atom are sparsely populated, and a state of population inversion is achieved between E_6 and E_5, E_3 levels and between E_4 and E_3 levels.
- Random photons emitted spontaneously prompt stimulated emission and lasing occurs. The transition E_6 and E_3 generates a laser beam of red colour of wavelength 6328Å. Other possible transitions produce 3.39 μm and 1.15 μm laser beam respectively as shown in fig.
- From the level E_3 the neon atoms drop to E_2 level spontaneously. E_2 level is however a metastable state. Consequently, neon atoms tend to accumulate at E_2 level. It is necessary that these atoms are brought to the ground state E_1 quickly; otherwise the number of atoms at the ground state will go on diminishing and the laser ceases to function.
- If the discharge tube is made narrow, the probability of atomic collisions with the tube walls increases. Because of frequent collisions with the walls, the neon atoms rapidly drop to the ground level and will be available for excitation once again.
- He-Ne laser operates in continuous wave mode and is widely used. It is widely used in laser printing, bar code reading, etc.

Advantages

- He-Ne laser operates in continuous wave mode.
- No separate cooling system is required.
- Optical properties are well defined

Limitations

- But we get very low output power

Engineering Application of Laser

1) It is used for cutting, drilling in metal industries
2) It is used for cutting and polishing of diamond
3)

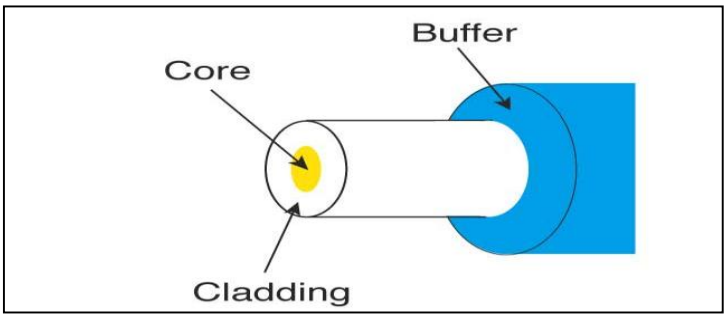
Fiber Optics

An optical fiber is a thin transparent thread or fiber made up of glass or clear plastic as thin as human hair used to transmit light signal along its length.

Structure of optical fibre

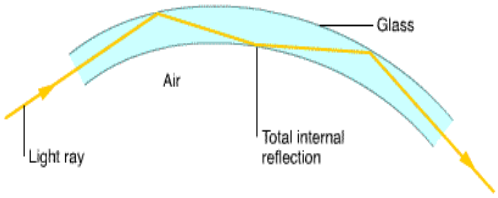
- **Core** – Inner most light guiding region
- **Cladding** – coaxial middle region having refractive index less than core

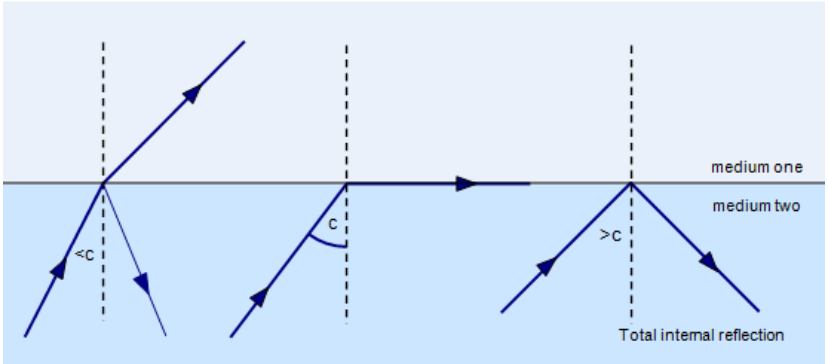
- **Buffer coating (Sheath)** - Plastic coating that protects the fiber from damage and moisture.
- Refractive index n_1 of the core is always greater than the Refractive Index n_2 of the cladding.



Propagation of light in an Optical Fiber

Optical Fiber works on the principle of **Total Internal Reflection**.





When a light ray travel from denser medium to rarer medium it bends away from the normal. As the angle of incidence is increased angle of refraction increases and at critical angle of incidence the ray is refracted by 90^0 and emerges along the boundary of the medium. If the angle of incidence is further increased the light is reflected back into the same medium. This is called as Total Internal Reflection.
The angle of incidence for which the angle of refraction is 90^0 is known as **critical angle ϕ_c**.

TIR takes place when a ray of light strikes the core-cladding boundary of an optical fiber at an angle larger than the critical angle with respect to the normal to the surface.

Then by Snell's Law at critical angle of incidence ϕ_c

$$n_1 \sin \phi_c = n_2 \sin r \qquad \text{Where } r = \text{angle of refraction}$$

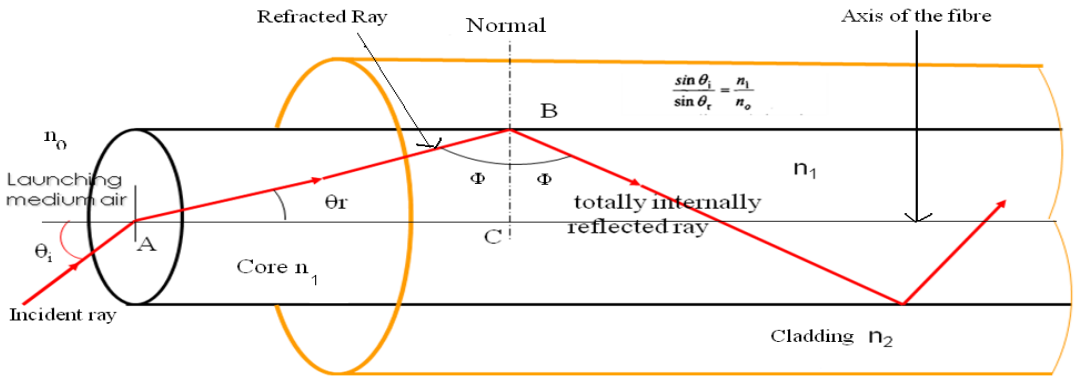
Or $$\frac{\sin \phi_c}{\sin r} = \frac{n_2}{n_1}$$

But at critical angle ϕ_c $\quad$ $\boldsymbol{r = 90^o}$

Therefore $$\frac{\sin \phi_c}{\sin 90} = \frac{n_2}{n_1} \qquad \text{or} \qquad \sin \phi_c = \frac{n_2}{n_1}$$

Acceptance Angle of an Optical fiber

Let a light is launched through an optical fiber of core refractive index n_1 and the cladding refractive index n_2 and let n_0 be the refractive index of the medium (air) through which the light is launched. Let the light ray enters the fiber at an angle θ_i to the axis of the fiber. The ray refracts at an angle θ_r and strikes the core cladding interface at an angle ϕ



Applying Snell's law to the launching face of the fiber,

$$n_0 \sin\theta_i = n_1 \sin\theta_r$$

From Δ ABC, $\sin\theta_r = \sin(90-\phi) = \cos\phi$

Therefore $n_0 \sin\theta_i = n_1 \cos\phi$

When $\phi = \phi_c$, $\theta_i = \theta_{i(max)} = \theta_0$ (Maximum Acceptance angle)

Or $n_0 \sin\theta_0 = n_1 \cos\phi_c$

Or $n_0 \sin\theta_0 = n_1\sqrt{1-\sin^2\phi_c}$

But $\sin\phi_c = \frac{n_2}{n_1}$

Therefore $n_0 \sin\theta_0 = n_1\sqrt{1-\frac{n_2^2}{n_1^2}}$

Or $n_0 \sin\theta_0 = n_1\sqrt{\frac{n_1^2-n_2^2}{n_1^2}}$

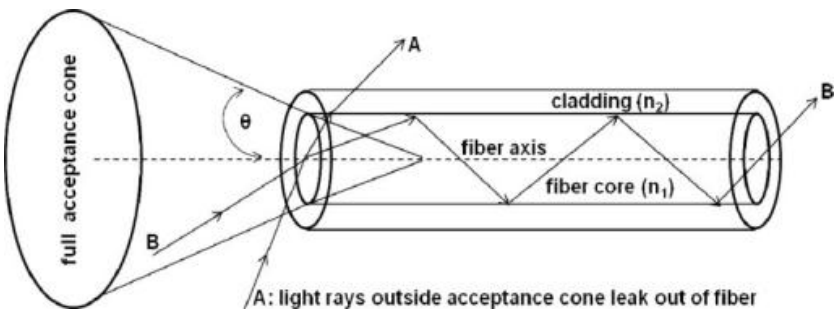
Or $n_0 \sin\theta_0 = \sqrt{n_1^2 - n_2^2}$

For air $n_0 = 1$ Therefore $\sin\theta_0 = \sqrt{n_1^2 - n_2^2}$

Or $\theta_0 = \sin^{-1}\sqrt{n_1^2 - n_2^2}$

Thus the Maximum Acceptance angle is defined as the angle for which the light is propagated through the Optical Fiber.

Acceptance cone: It is twice of the acceptance angle i.e. $2\theta_0$. The light ray within the core is accepted and transmitted through the Optical Fiber.



Numerical Aperture

Numerical Aperture is defined as the sin of the acceptance angle.

$$N.A. = \sin\theta_0 = \sqrt{{n_1}^2 - {n_2}^2}$$

Or $$N.A. = \sqrt{{n_1}^2 - {n_2}^2}$$

Fractional Refractive Index Change

Fractional refractive index change is the fractional difference between the refractive indices of the core and the cladding.

$$\Delta = \frac{n_1 - n_2}{n_1}$$

Numerical Aperture in terms of Δ

$$N.A. = \sqrt{{n_1}^2 - {n_2}^2}$$

Or $$N.A. = \sqrt{(n_1 - n_2)(n_1 + n_2)}$$

Or $$N.A. = \sqrt{2n_1\left(\frac{n_1 - n_2}{n_1}\right)\left(\frac{n_1 + n_2}{2}\right)}$$

We have $\Delta = \frac{n_1 - n_2}{n_1}$ and $\left(\frac{n_1 + n_2}{2}\right) \cong n_1$

Therefore $N.A. = \sqrt{2n_1\,\Delta\,n_1}$ or $N.A. = n_1\sqrt{2\,\Delta}$

Thus N.A. determines the light gathering power of an optical fiber. It is the measure of the amount of light accepted by the optical fiber.

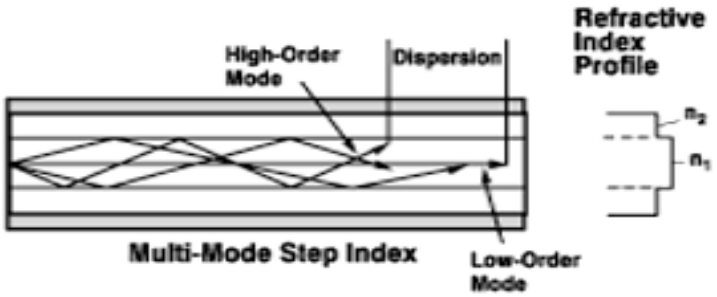
Modes of propagation

Modes are the possible number of paths of light that propagates through optical fibre.

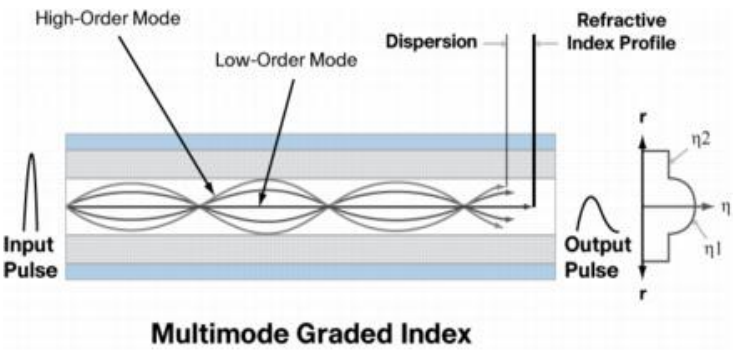
Single mode fibre: Single mode fiber has a core diameter of 8 to 12 µm. Allows only single beam of light to travel along the axis of optical fibre. N.A. and Acceptance angle are very small.

Single-Mode Step Index

Multimode Step-Index Fiber: Multimode fiber has a core diameter of 50 or 62.5 µm (sometimes even larger). Allows several light paths or *modes.* This causes *modal dispersion* – some modes takes longer time to pass through the fiber than others because they travel a longer distance.

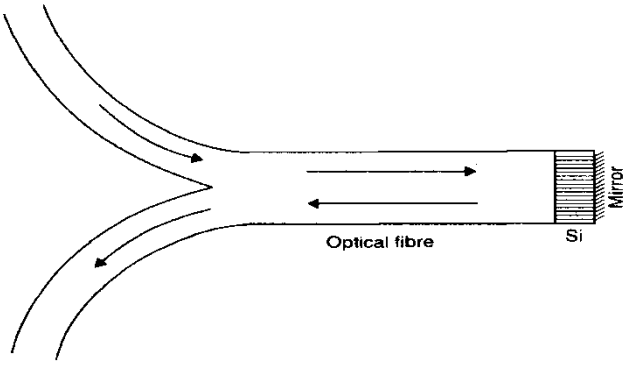


Multimode Graded-Index Fiber: The index of refraction gradually changes from core to cladding. Modes that travel further also move faster. This reduces *modal dispersion* so the bandwidth is greatly increased.



Application of fiber optics

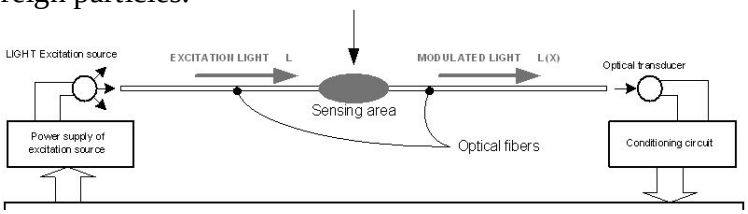
1) Temperature sensors



Temperature is measured by the modulation of intensity of the reflected light from a target, a silicon layer. The change in intensity of the reflected light is proportional to the change in temperature. The light is made incident from one of the branches of optical fiber which passes through the silicon layer. A mirror coating at the other end of the silicon layer reflects the light back which again travels through silicon layer and emerges out through another branch of the optical fiber and is collected by the photodetector.

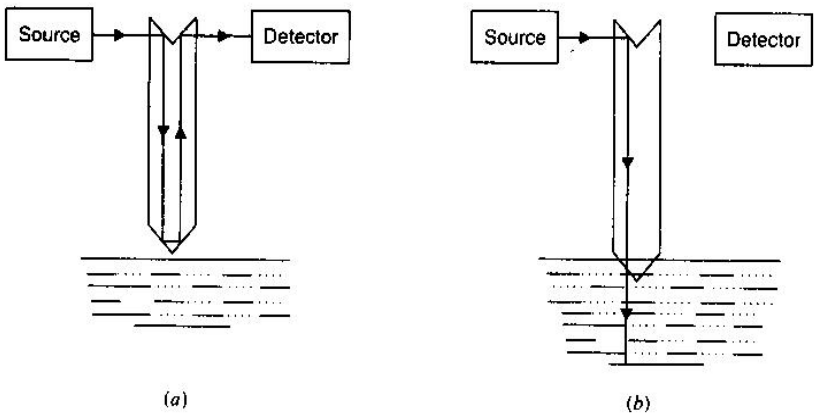
3) Smoke / Pollution Detector

A smoke and pollution detector can be built by using optical fibers. A beam of light radiating from one end of a fiber can be collected by another optical fiber. If foreign particles are present in the region between the two fibers, they scatter light. The variation in intensity of the light collected by the second optical fiber reveals the presence of foreign particles.



4) Liquid Level Detector

Principle: The liquid level detector described here is based on the principle of total internal reflection. A notch is made at one end of the optical fiber and the other end is chamfered. The light is made incident on one of the inclined faces of the notch. The light turns through 90º and travel through fiber. On reaching the chamfered end it gets internally reflected at the fiber-air boundary, if the liquid is below the desired level. Then it again turns through 90º and is detected at the detector. When the liquid rises and touches the fiber end the light is transmitted through the liquid. Hence the photodetector does not receive light thus indicating the liquid level.



1. Calculate the refractive indices of the core and cladding material of a fibre having NA=0.22 and Δ=0.012. (4)
2. In Newton's ring experiment the diameter of the 15th ring was found to be 0.59cm and that of the 5th ring was 0.336cm. if the radius of the Plano-convex lens is 100cm.Calculate the wavelength of light used. (04)

3. Calculate the specific rotation if the plane of polarization is turned through 26.4°, traversing 20cm length of 20% sugar solution. (04)
4. If the acceptance angle for a given fibre is 68.16°. Calculate the maximum entrance angle and numerical aperture. If the cladding glass has a R.I. 1.52 calculate the refractive index (R.I.) of the core glass. (04)

5. For a glass plate of R.I.(=1.54) Calculate the angle of polarization and also angle of refraction.(04)

6. He-Ne Laser is emitting a laser beam with an average power of 4.5mW. Find the number of photons emitted per second by the laser. The wavelength emitted is 6328 A°. (04)

7. Calculate acceptance angle for an optical fibre whose core R.I. is 1.48 and cladding R.I.is 1.39 (04)

8. Light of wavelength 5500 Å falls normally on a thin wedge shaped film of R.I. 1.4 forming fringes that are 2.5 mm apart. Find the angle of wedge. (3)

9. The numerical aperture of an optical fiber is 0.5 and Core refractive index is 1.54. Find the refractive index of the cladding. (3)
10. Newton's rings are observed in the reflected light of wavelength 5900Å. The diameter of tenth dark ring is 0.5 cm . Find the radius of curvature of lens used. (4)

11. Calculate the specific rotation of the sugar solution if the plane of polarisation of plane polarised light is rotated through 6.5^0 in passing through a length of 2.0 dm of sugar solution of 5% concentration. (4)

12. Calculate the refractive indices of the core and cladding material of a fibre having NA=0.22 and Δ=0.012. (4)

13. A fibre cable has an acceptance angle of 30° and a core refractive index 1.4. Calculate the refractive index of cladding.

14. Light of wavelength 5880 Å is incident on a thin film of glass (μ=1.5) such that the angle of refraction in the plate is 60°. Calculate the smallest thickness of plate which will make it dark by reflection.

15. Calculate the numerical aperture and acceptance angle of optical fibre of refractive index for core and cladding are 1.62 and 1.52 respectively.